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Global–local integrated path planning strategy for agricultural robots in complex hilly orchard environments

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Hilly orchards, which account for over 60% of China's orchard area, present complex terrain that limits the effectiveness of conventional agricultural machinery, resulting in high labor requirements and low operational efficiency. Agricultural robots offer a promising solution, with path planning as a critical enabling technology. However, existing methods often suffer from limited terrain adaptability, inadequate obstacle avoidance, poor dynamic responsiveness, and compromised operational safety. To address these challenges, this study proposes a dual closed-loop global–local path planning framework for orchard robots operating in hilly environments. A terrain-aware traffic cost model and a slip-inclusive kinematic model are first developed to characterize traversability and motion stability. Building on these models, the A* algorithm is enhanced for global path planning, while the Dynamic Window Approach (DWA) is improved to enable local obstacle avoidance while satisfying terrain stability constraints. A double closed-loop control architecture integrates global path constraints with local environmental feedback and real-time replanning, achieving a balance between global optimality and local responsiveness. Field experiments demonstrate that the proposed strategy reduces path length by 2.1–19.8% and slip ratio by 2.7–8.0% compared with conventional methods, achieves a 100% obstacle-avoidance success rate, and maintains lateral trajectory deviations below 15.2 cm. These results validate the method as a safe, stable, and efficient approach for robotic operations in hilly orchards, highlighting its potential for practical deployment in complex agricultural environments.

KEYWORDS
agricultural robot, global–local fusion, hilly orchard, improved A* algorithm, improved dynamic window approach, path planning

1 Introduction

China is the world's largest fruit producer, with an orchard planting area exceeding 12 million hectares. More than 60% of these orchards are located in the red-yellow soil hilly regions of southern China and the loess plateau regions of northern China (Zhang et al., 2024; Xie et al., 2026). These orchards are typically characterized by complex terrain conditions, including slopes ranging from 5° to 30°, fragmented field layouts, narrow access roads, and irregular tree spacing (Kumar et al., 2021). Under such conditions, the deployment of large-scale wheeled agricultural machinery is severely constrained, and key operations such as plant protection, harvesting, and transportation are still predominantly performed manually (Jiang

et al., 2025). According to survey data from the Ministry of Agriculture and Rural Affairs, labor costs in hilly orchards account for more than 65% of total production expenses, while more than half of fruit farmers are over 60 years old. Consequently, the labor shortage and the aging rural workforce have become critical bottlenecks, limiting the sustainable development of China's fruit industry (Wang et al., 2021; Khatri and Shrestha, 2022; Tong et al., 2024).

To address these challenges, agricultural robots have emerged as a promising solution due to their autonomy and adaptability in unstructured environments (Zhang et al., 2019). Among the core technologies enabling autonomous operation, path planning plays a fundamental role, acting as the "decision-making brain" of agricultural robots by directly influencing navigation efficiency, operational safety, and task performance (Chakraborty et al., 2022; Kawas, 2025; Chandrayan and Alvarez, 2026). Consequently, extensive research has been conducted on robot path planning in agricultural environments (Shen et al., 2025a; Gao Y. et al., 2026; Gao L. et al., 2026). Classical global planning methods such as A* and Rapidly-exploring Random Trees (RRT) are widely adopted for generating feasible routes in orchard environments. However, these methods typically rely on prior map information and exhibit limited adaptability to dynamic obstacles and unstructured terrain variations (Ye et al., 2023; Li et al., 2025; Yuan and Luo, 2025). At the local planning level, methods such as the Artificial Potential Field (APF) and Dynamic Window Approach (DWA) enable real-time obstacle avoidance (Kobayashi and Motoi, 2022; Thanh et al., 2025), but they are prone to local minima, trajectory deviation, and a lack of global optimality guarantees (Lu and Da, 2025; Yu et al., 2025). As a result, traditional planning frameworks often struggle to simultaneously ensure global optimality, local safety, and terrain adaptability in complex orchard environments.

To further enhance path planning performance, bio-inspired and evolutionary optimization algorithms have been increasingly incorporated into agricultural robot navigation. A Dynamic Genetic Algorithm–Ant Colony Optimization (DGA-ACO) method to improve global path optimality and convergence efficiency (Zhang et al., 2025). Similarly, extending the application of bio-inspired optimization, an improved Jellyfish Search (JS) algorithm for multi-robot collaborative task planning, demonstrating enhanced optimization and scheduling performance in complex vertical farming environments (Shen et al., 2025b). In recent years, deep reinforcement learning (DRL) has shown strong potential for robot navigation in complex, unstructured environments. Representative studies include end-to-end policy learning for obstacle avoidance (Gao Y. et al., 2026; Gao L. et al., 2026) and adaptive trajectory planning in dynamic agricultural scenarios (Martin and Collins, 2026). In addition, learning-based navigation approaches, such as Bayesian deep learning for terrain traversability estimation, have been applied to off-road robotic systems (Bhosale et al., 2026). Despite their promising performance, these methods generally require large-scale training data, may lack explicit stability and safety guarantees, and often do not incorporate physically interpretable terrain cost formulations. In contrast, model-based approaches provide deterministic behavior and explicitly incorporate physical constraints. However, most existing model-based methods insufficiently address adaptive decision-making in highly dynamic environments. Recent studies have further expanded research on agricultural robot path planning. Bio-inspired optimization methods, such as the improved sand cat swarm optimization (ISCSO) algorithm, have been shown to enhance convergence performance and path optimality in complex agricultural scenarios (Wang et al., 2025). Meanwhile, continuous-curvature planning strategies have been proposed to improve motion smoothness and feasibility during headland operations, enabling safer and more practical field navigation (Mier et al., 2025). From a system-level perspective, integrated planning frameworks combining farm management information systems (FMIS), cloud-based route planning, and unmanned ground vehicles (UGVs) have demonstrated improvements in operational efficiency and decision support capabilities (Asiminari et al., 2024). Furthermore, comprehensive surveys have classified existing methods into classical algorithms, bio-inspired optimization, sampling-based planning, and machine learning-based approaches, highlighting a growing trend toward hybrid and intelligent planning frameworks (Gao Y. et al., 2026; Gao L. et al., 2026).

Recent research has increasingly focused on multi-objective and intelligent path planning. AgriPath, a multi-objective framework for dynamic agricultural environments (Yang et al., 2025), and the CBS-HT algorithm to improve safe coordination among heterogeneous agricultural robots (Jo and Son, 2025). Katikaridis et al. (2026) introduced a centralized UAV–UGV cooperative planning framework, and Zhang et al. (2026) integrated path planning and trajectory tracking for orchard inspection robots. Although these methods improve navigation efficiency and coordination, they generally do not explicitly address terrain-induced mobility constraints, such as wheel slip and stability, in hilly orchard environments. Despite these advances, path planning in hilly orchard environments remains particularly challenging due to the coexistence of steep slopes, uneven terrain, soil slipperiness, and dynamic obstacles. Existing methods typically focus on individual aspects such as path optimality, coverage efficiency, or system integration, while neglecting the coupled requirements of terrain adaptability, motion stability, real-time replanning, and safe navigation under sloped conditions (Wang et al., 2025; Gao Y. et al., 2026; Gao L. et al., 2026). In addition, many integrated planning frameworks adopt a loosely coupled "global planning + local tracking" structure, which lacks sufficient feedback mechanisms to handle real-time environmental changes (You et al., 2026). This often leads to suboptimal detours and failure to return to the intended operational path after local obstacle avoidance. Moreover, existing approaches rarely incorporate orchard-specific operational constraints, such as fruit tree protection zones, slip risk, and slope-dependent stability constraints, limiting their practical applicability. To address these limitations, this paper proposes a terrain-aware, closed-loop path planning framework for agricultural robots operating in hilly orchards. The main contributions are summarized as follows: (i) The terrain traffic cost model (slope, roughness, safety buffer) adapts static cost mapping to hilly orchards, which is a scenario-specific extension of standard grid-based cost maps. (ii) The slip-aware kinematic model modifies conventional discrete-time robot kinematics using field-fitted slip ratios. (iii) The improved A* and DWA algorithms incorporate these models into classic frameworks. Genuinely novel elements (not present in prior A*-DWA, PSO-based, or terrain-aware agricultural planning): (i) First double closed-loop architecture with environment-triggered and path-deviation-triggered global replanning for hilly orchards, previous integrations (Fan et al., 2025; Dong et al., 2026) use open-loop "global + local" without feedback replanning. (ii) Unified evaluation function in DWA that simultaneously optimizes terrain stability (roll/pitch/slip), global path tracking, and Kalman-filter-based dynamic obstacle prediction; no prior work combines all three in a single local planner for sloping orchards. (iii) PSO-tuned weights specifically for the

trade-off among tracking accuracy, slip reduction, and dynamic avoidance in 5°–25° slopes. Table 1 compares our method with three representative studies, highlighting the novel closed-loop replanning and slip-terrain co-optimization absent in others.

2 Materials and methods

2.1 Overall architecture of the path planning system for agricultural robots in hilly orchards

The path planning system designed in this paper adopts a hierarchical architecture, which is divided into 5 core modules: environment perception layer, global path planning layer, local dynamic planning layer, integrated decision-making layer, and motion control layer.

2.1.1 Composition of the hardware system

The hardware system serves as the carrier for the path planning strategy. Given the topographic characteristics of hilly orchards, a tracked mobile robot chassis is selected to meet the driving demands of steep slopes and soft ground. The core hardware includes:

(1) *Perception module*: A 16-line LiDAR is used to complete environmental obstacle and terrain scanning, and a monocular vision camera is used to realize dynamic obstacle recognition. RTK-GNSS combined with an IMU provides centimeter-level positioning, and a dual-axis tilt sensor measures the robot's pitch and roll angles in real time, enabling terrain adaptability (Duan et al., 2024; Melvin et al., 2026).
(2) *Decision and control module*: NVIDIA Jetson Xavier NX is used as the main controller to complete environmental data processing, path planning algorithm operation, and decision

TABLE 1 Comparison with closely related path planning studies for agricultural robots.

Feature	Improved A* + DWA (Yu et al., 2025)	AgriPath (Yang et al., 2025)	DGA-ACO (Zhang et al., 2025)	Improved JS (Shen et al., 2025a,b)	CBS-HT (Jo and Son, 2025)	Orchard inspection (Zhang et al., 2026)	UAV–UGV coordination (Katikaridis et al., 2026)	Our strategy
Global path optimization	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local obstacle avoidance	Yes (DWA)	Yes	Partial	Partial	Yes	Yes	Yes	Yes (Improved DWA)
Terrain cost (slope and roughness)	Partial	Partial	No	No	No	Partial	Partial	Yes
Slip kinematic model	No	No	No	No	No	No	No	Yes
Terrain stability evaluation	No	Partial	No	No	No	Partial	No	Yes (roll, pitch, and slip)
Dynamic obstacle prediction	No	Yes	Partial	No	No	Partial	Partial	Yes (Kalman filter)
Closed-loop global–local replanning	No	Yes	No	No	No	Partial	Yes	Yes (dual-trigger)
Fruit-tree safety constraint	No	No	No	No	No	Yes	No	Yes
Multi-robot collaboration	No	No	No	Yes	Yes	No	Yes	No
Real orchard validation	Yes	Field	Simulation	Vertical farm	Simulation	Yes	Semi-structured farmland	Yes (5°–25° hilly orchard)
Validation with dynamic obstacles	Limited	Yes	Simulation	Simulation	Yes	Limited	Yes	Yes

The abbreviations are: A*, A-star algorithm; DWA, dynamic window approach; DGA-ACO, dynamic genetic algorithm–ant colony optimization; JS, Jellyfish Search algorithm; CBS-HT, Conflict-Based Search for Heterogeneous Teams; UAV, Unmanned Aerial Vehicle; UGV, Unmanned Ground Vehicle; Kalman, Kalman filter-based state estimation.

instruction output. It is matched with the STM32 bottom controller to complete motor drive and motion state feedback.

(3) Actuation module: Dual DC servo motors are used to drive the left and right tracks, and a reducer is used to achieve high torque output, which adapts to the climbing demand of hilly terrain. The maximum climbing angle can reach 30°, and the rated driving speed is 0–1.2 m/s.

2.1.2 Hierarchical logic of the software system

(1) *Environment perception layer*: It is responsible for collecting field environment data, completing grid map construction, obstacle identification and classification, dynamic obstacle motion prediction, and robot pose and motion state collection, providing environment and state input for upper and lower layer planning.

(2) *Global path planning layer*: Based on the prior grid map, taking the operation start point and end point as input, combined with the traffic cost model of hilly orchards, the improved A* algorithm is used to generate the global optimal reference path, ensuring the global optimality, terrain passability, and operation standardization of the path.

(3) *Local dynamic planning layer*: Based on the real-time perceived environment data, taking the global path as the tracking target, the improved DWA algorithm is used to generate the local optimal trajectory, complete dynamic obstacle avoidance and local terrain mutation adaptation, and ensure the real-time performance and safety of robot driving.

(4) *Integrated decision-making layer*: It is the core connection module of the system, responsible for realizing the double closed-loop integration of global and local planning. On the one hand, it discretizes the global path into tracking target points and sets constraint boundaries for local planning. On the other hand, it collects real-time state feedback from local planning. It triggers global replanning when environmental or path deviations exceed the threshold, enabling dynamic adaptation between global and local planning.

(5) *Motion control layer*: It receives the optimal trajectory output by the integrated decision-making layer, calculates the speed instructions of the left and right tracks based on the robot kinematic model, completes trajectory tracking control, and feeds back the motion state of the robot in real time to form a bottom control closed loop.

This architecture strictly follows the logic of "global overall planning - local adaptation - integrated decision-making - closed-loop feedback," ensuring the global optimality of the path, enabling real-time response to complex dynamic environments, and fully compatible with the unstructured operational scenarios of hilly orchards.

2.2 Modeling of complex hilly orchard environment and discrete-time domain model of agricultural robots

Accurate environment and robot modeling are the basis for the path-planning strategy. This paper develops a grid-based traffic cost model for hilly orchards and a discrete-time slip kinematic model for tracked robots, both of which fit the actual field working conditions.

2.2.1 Grid-based modeling of complex hilly orchard environment

To accurately characterize the complex terrain features of hilly orchards, a grid-based environmental modeling approach was adopted. The operational area was discretized into uniform grid cells with a spatial resolution of 0.2 m × 0.2 m. Each grid cell stores four categories of essential terrain information, including elevation, slope, ground roughness, and obstacle attributes. The resulting grid map can be expressed as:

$$M = \left\{ G\left(i,j\right) \middle| i = 1,2,\ldots,m;\ j = 1,2,\ldots,n \right\}$$

where $G\left(i,j\right)$ denotes the grid cell located at the i-th row and j-th column, while m and n represent the total number of rows and columns in the grid map, respectively.

2.2.2 Calculation model of terrain parameters

Considering the significant terrain undulations commonly encountered in hilly orchard environments, two key terrain parameters, slope and ground roughness, were derived from the elevation data of the grid map. These parameters serve as fundamental indicators for terrain passability assessment.

(1) *Slope calculation*: Slope is a critical factor influencing the traversability of a terrain cell and directly affects the mobility performance of field robots (Yousefi et al., 2026). In this study, the slope of each grid cell was computed using a 3 × 3 neighborhood window and a finite-difference approximation of the terrain gradient. The calculation is expressed as:

$$\begin{cases} \mathrm{p} = \dfrac{\left[\mathrm{z}\left(\mathrm{i}+1,\mathrm{j}\right)+2\mathrm{z}\left(\mathrm{i}+1,\mathrm{j}\right)+\mathrm{z}\left(\mathrm{i}+1,\mathrm{j}+1\right)\right]-\left[\mathrm{z}\left(\mathrm{i}-1,\mathrm{j}\right)+2\mathrm{z}\left(\mathrm{i}-1,\mathrm{j}\right)+\mathrm{z}\left(\mathrm{i}-1,\mathrm{j}+1\right)\right]}{8\Delta d} \\ \mathrm{q} = \dfrac{\left[\mathrm{z}\left(\mathrm{i},\mathrm{j}+1\right)+2\mathrm{z}\left(\mathrm{i},\mathrm{j}+1\right)+\mathrm{z}\left(\mathrm{i}+1,\mathrm{j}+1\right)\right]-\left[\mathrm{z}\left(\mathrm{i},\mathrm{j}-1\right)+2\mathrm{z}\left(\mathrm{i},\mathrm{j}-1\right)+\mathrm{z}\left(\mathrm{i}+1,\mathrm{j}-1\right)\right]}{8\Delta d} \\ \grave{\mathrm{e}}\left(\mathrm{i},\mathrm{j}\right) = \arctan\left(\sqrt{\mathrm{p}^{2}+\mathrm{q}^{2}}\right) \end{cases} \quad (1)$$

where $z\left(i,j\right)$ represents the elevation value of the grid cell $G\left(i,j\right)$, Äd denotes the grid resolution, and p and q are the elevation gradients in the x- and y-directions, respectively. The resulting $\theta\left(i,j\right)$ represents the slope angle of the grid cell in radians.

(2) *Ground roughness calculation*: Ground roughness reflects local terrain irregularities and is an important indicator of vehicle stability and slip risk during robot navigation. In this study, terrain roughness was quantified using the standard

deviation of elevation values within a 3×3 neighborhood window:

$$\sigma(i,j) = \sqrt{\frac{1}{9}\sum_{k=i-1}^{i+1}\sum_{l=j-1}^{j+1}\left[z(k,l) - \overline{z}(i,j)\right]^2} \quad (2)$$

where $\sigma(i,j)$ denotes the roughness of the grid cell $G(i,j)$, and $\overline{z}(i,j)$ is the mean elevation of the corresponding 3×3 neighborhood. Larger values of $\sigma(i,j)$ indicate greater terrain unevenness, which may adversely affect robot stability and traversability.

This formulation enables a quantitative representation of local terrain characteristics and provides a reliable basis for subsequent passability evaluation and path-planning processes in complex hilly orchard environments.

2.2.3 Grid traffic cost model

To satisfy the navigation requirements of tracked agricultural robots operating in complex, hilly orchard environments, a multifactor weighted grid-traversal cost model was established. The model integrates terrain characteristics and environmental constraints by considering four key factors: slope cost, ground roughness cost, obstacle cost, and fruit-tree safety distance cost. The total traversal cost of a grid cell is defined as:

$$C(i,j) = \omega_1 C_\theta(i,j) + \omega_2 C_\sigma(i,j) + \omega_3 C_{obs}(i,j) + \omega_4 C_{safe}(i,j) \quad (3)$$

where $C(i,j)$ denotes the total traversal cost of the grid cell $G(i,j)$; $C_\theta(i,j)$, $C_\sigma(i,j)$, $C_{obs}(i,j)$, and $C_{safe}(i,j)$ represent the slope cost, roughness cost, obstacle cost, and safety distance cost, respectively. The weighting coefficients ω_1, ω_2, ω_3, and ω_4 satisfy $\omega 1 + \omega 2 + \omega 3 + \omega 4 = 1$. Based on preliminary field experiments and empirical parameter tuning, the weighting coefficients were set to $\omega_1 = 0.4$, $\omega_2 = 0.2$, $\omega_3 = 0.3$, and $\omega_4 = 0.1$.

The individual cost components are defined as follows.

(1) *Slope cost* $C_\theta(i,j)$: When the slope $\theta > 30^\circ$ (the maximum climbing angle of the robot), $C_\theta = +\infty$, which is an impassable grid; when $\theta \le 30^\circ$, $C_\theta = \theta / 30^\circ$, and the cost increases linearly with the increase in slope (Yousefi et al., 2026).

(2) *Roughness cost* $C_\sigma(i,j)$: When $\sigma > 0.15\text{m}, C_\sigma = +\infty$, which is an impassable grid; when $\sigma \le 0.15\text{m}, C_\sigma = \sigma / 0.15\text{m}$, and the cost increases linearly with the increase in roughness.

(3) *Obstacle cost* $C_{obs}(i,j)$: If there are static obstacles (fruit tree trunks, ditches, soil slopes) in the grid, $C_{obs} = +\infty$, which is an impassable grid; when there is no obstacle, $C_{obs} = 0$.

(4) Fruit tree safety distance cost $C_{safe}(i,j)$ to avoid the robot scraping the fruit trees, the area within 0.5 m of the fruit tree trunks is designated as a safety buffer zone. The cost of the grid in the buffer zone increases linearly with decreasing distance. The formula is $C_{safe} = 1 - d / 0.5\text{m}$, where d is the distance from the grid to the nearest fruit tree trunk, $d \le 0.5\text{m}$; when $d > 0.5\text{m}, C_{safe} = 0$.

2.2.4 Discrete-time slip kinematic model of tracked agricultural robots

Given the soft ground and easy slipping in hilly orchards, the traditional ideal non-slip kinematic model is abandoned, and a discrete-time slip kinematic model accounting for track slip is constructed to provide an accurate basis for path planning and trajectory tracking (Liu et al., 2025).

The tracked mobile robot adopts a differential drive mode, and the speeds of the left and right tracks are v_l and v_r respectively. In an ideal state, the linear velocity v and angular velocity ω of the robot are:

$$\begin{cases} v = \dfrac{v_r + v_l}{2} \\ \omega = \dfrac{v_r - v_l}{B} \end{cases} \quad (4)$$

where B is the track gauge of the robot's left and right tracks.

When driving on the soft ground of hilly orchards, the track will produce longitudinal slip. The longitudinal slip ratio σ is defined as the relative difference between the theoretical speed and the actual speed of the track, and the formula is: as follows:

$$\sigma = 1 - \frac{v_{real}}{v_{theo}} \quad (5)$$

where v_{real} is the actual driving speed of the track, and v_{theo} is the theoretical driving speed of the track. The slip ratio ó is positively correlated with the ground roughness and slope. The above empirical slip ratio equation $\sigma = 0.08\theta + 0.6\rho$ was fitted under specific experimental conditions: the test site was a hilly citrus orchard in Changsha, Hunan Province, with red-yellow soil, moderate moisture content (volumetric water content approx. 20–25%), slopes ranging from 5° to 25°, and roughness values between 0 and 0.15 m. The core assumptions of this model are: ① steady-state slip between the tracks and the ground (transient transitions are ignored); ② lateral slip is neglected; ③ the slip ratio has a linear relationship with slope and roughness, and the two factors are independent of each other. Generalisability limitations: When the soil type (e.g., sandy loam on the Loess Plateau), moisture content (> 30% after rain or < 15% during drought), orchard layout (irregular row/plant spacing), or weather conditions change, the coefficients 0.08 and 0.6 may vary significantly. Recalibration method: It is recommended to collect at least 10 sets of actual slip-ratio data across different slope-roughness combinations in the new environment and re-fit the coefficients using multiple linear regression. Coefficient sensitivity analysis: Taking a slope of 25° and a roughness of 0.12 m as an example, if the coefficient 0.08 changes by ±20%, the slip ratio changes by ±0.4%; if the coefficient 0.6 changes by ±20%, the slip ratio changes by ±1.44%. This indicates that the slip ratio is more sensitive to the roughness coefficient, so the roughness influence coefficient should be calibrated with higher priority.

After considering the slip correction, the actual linear velocity and angular velocity of the robot are:

$$\begin{cases} v_{real} = \dfrac{v_r(1-\sigma_r)+v_l(1-\sigma_l)}{2} \\ \omega_{real} = \dfrac{v_r(1-\sigma_r)-v_l(1-\sigma_l)}{B} \end{cases} \tag{6}$$

where σ_r and σ_l are the longitudinal slip ratios of the right and left tracks, respectively.

The pose of the robot can be expressed as $X=[x,y,\varphi]^T$, where x and y are the plane coordinates of the robot in the geodetic coordinate system, and o is the heading angle of the robot. Using discrete-time domain modeling, the control period is Δt, and the pose of the robot at time k is $X(k)=[x(k),y(k),\varphi(k)]^T$, then the pose update formula at time $k+1$ is:

$$\begin{cases} x(k+1)=x(k)+v_{real}(k)\cos\varphi(k)\cdot\Delta t \\ y(k+1)=y(k)+v_{real}(k)\sin\varphi(k)\cdot\Delta t \\ \varphi(k+1)=\varphi(k)+\omega_{real}(k)\cdot\Delta t \end{cases} \tag{7}$$

At the same time, to ensure the driving safety of the robot, the terrain stability constraint is set: the roll angle γ of the robot during driving must satisfy $|\gamma|\leq 15^\circ$ (critical rollover angle), and the pitch angle β must satisfy $|\beta|\leq 30^\circ$ (critical climbing angle), that is:

$$|\gamma(k)|\leq 15^\circ, \quad |\beta(k)|\leq 30^\circ \backslash \text{tag} \tag{8}$$

This discrete-time domain model fully accounts for the slip characteristics and terrain stability constraints of hilly orchards, is highly consistent with the robot's actual driving state, and provides accurate support for subsequent path planning (Fabregat-Jaen et al., 2025).

2.3 Design of global–local integrated path planning strategy

The core of the global–local integrated path planning strategy proposed in this paper is to realize the balance between global optimality and local real-time performance through a double closed-loop architecture. The whole strategy is divided into four core parts: global path-planning algorithm design, local dynamic obstacle-avoidance planning algorithm design, global–local integrated strategy architecture design, and strategy parameter optimization and stability analysis.

2.3.1 Global path planning algorithm: improved a* algorithm integrating terrain constraints

The traditional A* algorithm uses only Euclidean distance as the cost function, failing to account for terrain constraints in hilly orchards, resulting in paths with steep slopes, high slip risk, and proximity to fruit trees that cannot adapt to field operation requirements (Xu et al., 2024). In this study, the traditional A* algorithm is improved, and the grid traffic cost model constructed in Section 2.1 is integrated into the cost function to generate the global optimal path adapted to hilly terrain.

The core of the A* algorithm is the cost function $f(n)$, which is used to evaluate the priority of node n. The traditional cost function is:

$$f(n)=g(n)+h(n) \tag{9}$$

where $g(n)$ is the actual cost from the start point to the current node n, and $h(n)$ is the heuristic function from the current node n to the end point, that is, the estimated cost.

In this paper, the cost function is improved as follows:

(1) *Improvement of actual cost* $g(n)$: The traditional distance cumulative cost is abandoned, and the cumulative value of grid traffic cost is used as the actual cost. The formula is as follows:

$$g(n)=g(n_{parent})+C(n) \tag{10}$$

where $g(n_{parent})$ is the cumulative traffic cost from the start point to the parent node n_{parent}, and $C(n)$ is the traffic cost of the grid where the current node n is located, calculated by Equation 3.

(2) *Improvement of heuristic function* $h(n)$: The traditional Euclidean-distance heuristic does not account for terrain elevation variations, which may lead to an underestimation of the actual traversal cost and consequently reduce the search efficiency of the algorithm. In this paper, the improved Euclidean distance considering elevation difference is used as the heuristic function, and the formula is as follows:

$$h(n)=\lambda\cdot\sqrt{(x_n-x_{goal})^2+(y_n-y_{goal})^2}+\xi\cdot(z_n-z_{goal})^2 \tag{11}$$

where (x_n,y_n,z_n) is the coordinate of the current node n, $(x_{goal},y_{goal},z_{goal})$ is the coordinate of the end point, λ is the weight coefficient of the heuristic function, and ξ is the elevation influence coefficient. In this paper, $\lambda=1.2$ and $\xi=0.5$ are taken to ensure that the heuristic function satisfies admissibility and avoids the algorithm falling into a local optimum.

The flow of the improved A* algorithm is as follows: ① Initialize the open list Open and the closed list Close, and add the start node to the Open list. ② If the Open list is empty, the algorithm ends, and there is no feasible path; otherwise, take out the node with the smallest $f(n)$ in the Open list as the current node. ③ If the current node is the end point, backtrack to generate the global optimal path, and the algorithm ends; otherwise, move the current node into the Close list. ④ Traverse the 8 neighborhood nodes of the current node. If the neighborhood node is an impassable grid or already in the Close list, skip it; otherwise, calculate $f(n),g(n),h(n)$ of the neighborhood node. ⑤ If the neighborhood node is not in the Open list, add it to the Open list and set the current node as its parent node; if it is already in the Open list, compare the new $g(n)$ with the original $g(n)$, and update the cost and parent node if the new value is smaller.

⑥ Repeat steps 2–5 until the endpoint is found or the Open list is empty.

Through the above improvements, the algorithm-generated global path fully considers constraints such as slope, roughness, and fruit tree safety distance in hilly orchards, ensuring the path's passability, stability, and global optimality, and providing a reliable benchmark for local planning (Huang et al., 2025).

2.3.2 Local dynamic obstacle avoidance planning algorithm: improved DWA algorithm

The traditional DWA algorithm only considers obstacle avoidance and speed constraints, without accounting for terrain stability, slip risk, and global path-tracking requirements in hilly orchards. It is easy to have problems such as deviation from the global path after local obstacle avoidance, unstable driving, slipping, and rollover. In this paper, the traditional DWA algorithm is improved by integrating dynamic obstacle prediction, terrain stability constraints, and global path-tracking constraints to enable real-time optimization of local paths.

1. Basic principle of DWA algorithm

The core of the DWA algorithm is to sample multiple sets of speeds in the speed space (v,ω) of the robot, predict the driving trajectory of each set of speeds in the future time window, score all trajectories through the evaluation function, select the optimal trajectory with the highest score, and output the corresponding speed command (Kumar et al., 2021).

The constraints of the speed space mainly include: ① Maximum/minimum speed constraints of the robot: $v\in[v_{min},v_{max}],\omega\in[\omega_{min},\omega_{max}]$. ② Motor acceleration and deceleration constraints: $v\in[v_c-a_{max}\Delta t,v_c+a_{max}\Delta t],\omega\in[\omega_c-\alpha_{max}\Delta t,\omega_c+\alpha_{max}\Delta t]$, where v_c and $\grave{u}_c$ are the current speeds, and a_{max} and $\acute{a}_{max}$ are the maximum linear acceleration and angular acceleration. ③ Safety obstacle avoidance constraint: $v\le\sqrt{2\cdot dist(v,\omega)\cdot a_{max}}$, where $dist(v,\omega)$ is the distance from the predicted trajectory to the nearest obstacle.

2. Terrain-adaptive improved DWA algorithm

In this paper, the traditional DWA algorithm is improved from three aspects:

(1) *Dynamic obstacle motion prediction*: Aiming at dynamic obstacles (fruit farmers, poultry) in hilly orchards, the Kalman filter algorithm is used to predict their motion trajectories in the next 3 s, and the predicted position is incorporated into the obstacle avoidance constraint to avoid the "lagging obstacle avoidance" problem of the traditional algorithm. The prediction and update formulas of the Kalman filter are as follows:

$$\begin{cases}\hat{X}(k|k-1)=A\hat{X}(k-1|k-1)+Bu(k)\\ P(k|k-1)=AP(k-1|k-1)A^T+Q\\ K(k)=P(k|k-1)H^T\left[HP(k|k-1)H^T+R\right]^{-1}\\ \hat{X}(k|k)=\hat{X}(k|k-1)+K(k)\left[Z(k)-H\hat{X}(k|k-1)\right]\\ P(k|k)=\left[I-K(k)H\right]P(k|k-1)\end{cases}\tag{12}$$

where $\hat{X}$ is the state vector (position, velocity) of the obstacle, A is the state transition matrix, B is the control matrix, P is the covariance matrix, K is the Kalman gain, H is the observation matrix, Q and R are the covariance matrices of process noise and observation noise, respectively, and Z is the observation value.

(2) *Terrain constraint expansion of speed space*: Based on traditional speed constraints, slope and slip constraints are added. The maximum driving speed is dynamically adjusted based on the current driving slope to avoid overspeeding on downhill slopes and insufficient power on uphill slopes. The constraint formula is as follows:

$$v_{max}(\theta)=\begin{cases}1.2m/s, & \theta\le 5^\circ\\ 1.2-0.03\theta, & 5^\circ<\theta\le 25^\circ\\ 0.5m/s, & \theta>25^\circ\end{cases}\tag{13}$$

where θ is the current driving slope, positive for uphill and negative for downhill, and the absolute value is taken for calculation when going downhill.

(3) *Evaluation function optimization*: Based on the traditional evaluation function, the terrain stability cost item and the global path tracking cost item are added to construct a multi-objective optimization evaluation function. The formula is as follows:

$$J=\alpha\cdot J_{goal}+\beta\cdot J_{obs}+\gamma\cdot J_{stable}+\delta\cdot J_{track}\tag{14}$$

where J_{goal} is the target point heading cost, J_{obs} is the obstacle avoidance cost, J_{stable} is the terrain stability cost, J_{track} is the global path tracking cost; $\alpha,\beta,\gamma,\delta$ are the weight coefficients of each cost item, and satisfy $\alpha+\beta+\gamma+\delta=1$. All cost items are normalized to the range $\bar{0},1]$.

The calculation rules of each cost item are as follows: ① Target point heading cost J_{goal}: Evaluate the heading deviation between the end of the trajectory and the target point of the global path, $J_{goal}=1-|\Delta\varphi|/\pi$, where $\Delta\varphi$ is the heading deviation. The smaller the deviation, the higher the score. ② Obstacle avoidance cost J_{obs}: Evaluate the minimum distance between the trajectory and the obstacle, $J_{obs}=dist_{min}/d_{safe}$, where $dist_{min}$ is the minimum distance from the trajectory to the nearest obstacle, and d_{safe} is the safety distance (0.5 m in this paper). The farther the distance, the higher the score. ③ Terrain stability cost J_{stable}: Evaluate the driving stability of the trajectory, calculated based on the slope, roll angle, and slip ratio of the predicted trajectory, $J_{stable}=1-\left(\theta_{max}/30^\circ+||\gamma_{max}||/15^\circ+\sigma_{max}/0.2\right)/3$. The more stable the terrain, the higher the score. ④ Global path tracking cost J_{track}: Evaluate the lateral deviation between the trajectory and the global reference path, $J_{track}=1-d_{dev}/d_{max}$, where d_{dev} is the lateral deviation from the end of the trajectory to the global path, and d_{max} is the maximum allowable deviation (1 m in this paper). The smaller the deviation, the higher the score.

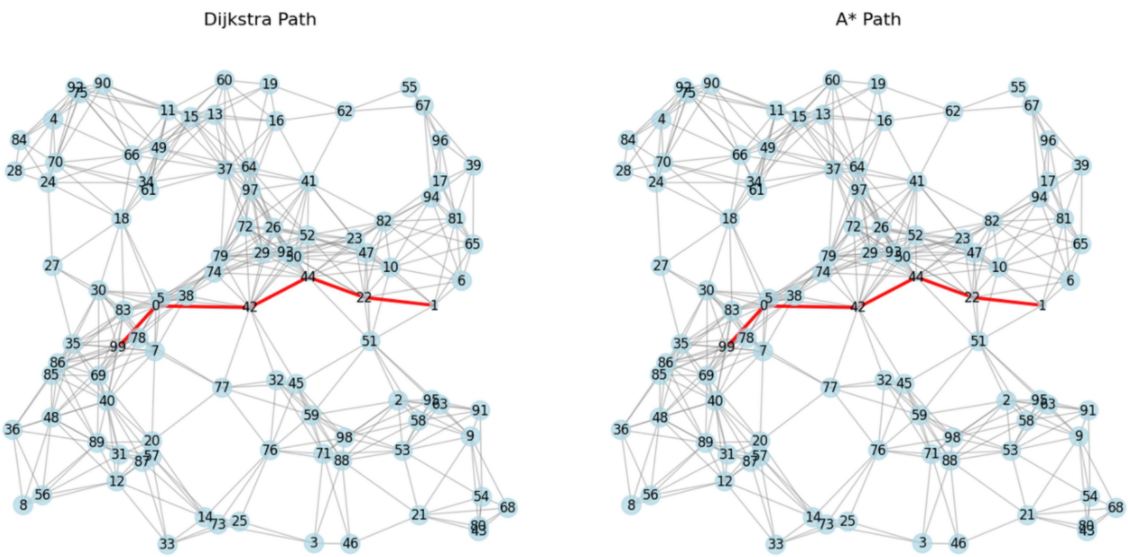


FIGURE 1
Design of global–local double closed-loop integrated strategy.

Through the above improvements, the DWA algorithm can not only realize real-time obstacle avoidance of dynamic obstacles, but also fully ensure the terrain stability of robot driving, and strictly track the global reference path at the same time, solving the problem of "deviation after obstacle avoidance" of the traditional algorithm, and fully adapting to the operation requirements of hilly orchards.

2.3.3 Design of global–local double closed-loop integrated strategy

Aiming at the problems of rigid connection and poor dynamic adaptability of the traditional "global + local" open-loop integrated strategy, this paper designs a double closed-loop integrated architecture of "global path constraint - local state feedback - dynamic replanning trigger," which realizes the deep integration of global planning and local planning. The core logic is shown in Figure 1.

1. Core architecture of the integrated strategy

The integrated strategy is divided into two closed loops: the main loop layer and the replanning trigger layer. The specific logic is as follows:

(a) Main loop layer: Local planning closed loop under global path constraints.

The core of the main loop layer is to accurately track local planning to the global path. The flow is as follows: ① Based on the operation start point and end point, the initial global optimal reference path $P_{global} = \{p_1, p_2, \ldots, p_N\}$ is generated by the improved A* algorithm, where $p_i = (x_i, y_i, \varphi_i)$ is the discrete path point of the global path. ② Based on the current pose of the robot, the preview point p_{target} on the global path is selected as the target point of local planning by the preview method, and the heading and position constraints are set for local planning. ③ The local planning module generates the local optimal trajectory P_{local} through the improved DWA algorithm based on the real-time environment data, taking the preview point as the target and the global path as the tracking benchmark, and outputs the speed command to the motion control layer. ④ The motion control layer completes trajectory tracking, feeds back the pose and motion state of the robot in real time, and enters the next main loop cycle.

(b) Replanning trigger layer: Global planning closed loop with local state feedback.

The core of the replanning trigger layer is to dynamically update the global path in response to changes in the field environment, and to avoid path failures caused by inconsistencies between the prior map and the actual environment. Two replanning trigger conditions are set in this paper, and global replanning is triggered when any one of the conditions is: met: ① Environmental deviation trigger: When the locally perceived unmarked obstacles continuously cover the global path for more than 3 m, or the elevation deviation between the local terrain and the prior map exceeds 0.3 m, it is determined that the prior map is inconsistent with the actual environment, and global replanning is triggered. ② Path deviation trigger: When the lateral deviation between the robot's actual driving trajectory and the global reference path continuously exceeds 1 m for more than 2 s, it is determined that the local planning has deviated from the global operation path, and global replanning is triggered.

The replanning flow is as follows: ① After triggering replanning, take the current pose of the robot as the new start point and the original operation end point as the target point, and correct the global grid map based on the updated local environment data. ② The improved A* algorithm is used to regenerate the global optimal path to replace the original global reference path. ③ The main loop layer continues to execute local planning and trajectory tracking based on the new global path to complete the closed-loop update.

(c) Execution Flow of the Integrated Strategy: The complete execution flow of the global–local integrated path planning strategy proposed in this paper is as follows:

(1) Before the operation, the environmental data of the hilly orchard is collected through LiDAR and RTK-GNSS, the

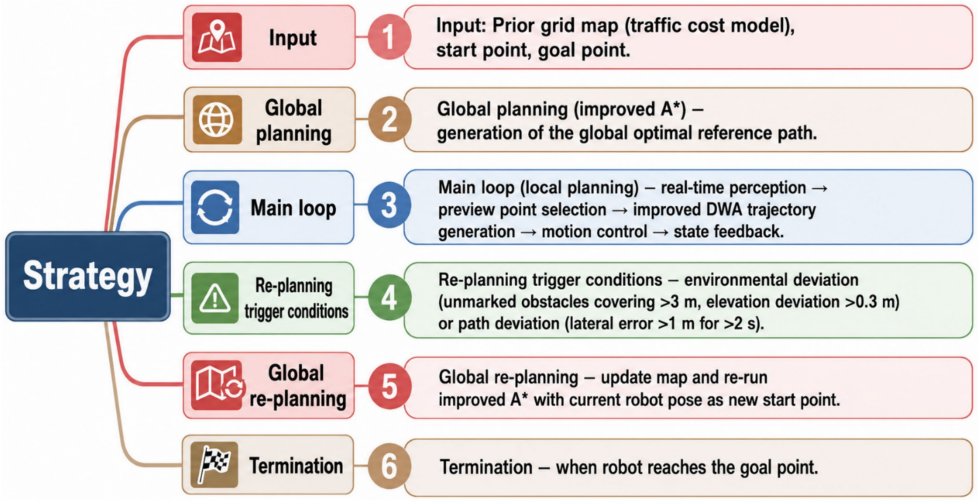


FIGURE 2
Flowchart of the proposed double closed-loop global–local integrated path planning strategy for hilly orchard robots.

prior grid map is constructed, and the operation start point and end point are set.

(2) The improved A* algorithm is used to generate the initial global optimal reference path, and the global path initialization is completed.

(3) The robot starts driving. The main loop layer collects the robot's pose and environment data in real time, selects the global path preview point, generates a local-optimal trajectory using the improved DWA algorithm, and controls the robot's drive.

(4) The replanning trigger conditions are detected in real time. If not met, continue executing the main loop; if met, trigger global replanning and update the global reference path.

(5) When the robot reaches the operation endpoint, the path planning flow ends.

This integrated architecture not only ensures the optimality of the global path and the standardization of operations but also enables real-time response to the local dynamic environment, addresses the open-loop defect of the traditional integrated strategy, and effectively adapts to the complex, changeable operational scenarios of hilly orchards (Figure 2).

2.4 Parameter optimization and stability analysis of the integrated strategy

2.4.1 Parameter optimization based on particle swarm optimization

The evaluation function of the improved DWA algorithm proposed in this paper contains 4 weight parameters: α, β, γ, and ä. Manual parameter tuning is difficult to achieve multi-objective optimization, so the Particle Swarm Optimization (PSO) algorithm is used to globally optimize the weight parameters to achieve comprehensive optimization of path tracking accuracy, obstacle avoidance performance, and driving stability (Ahmad et al., 2025; Jiang et al., 2026).

(1) *Optimization variable setting*: The 4 weight coefficients of the evaluation function are taken as optimization variables, that is, $X=[\alpha,\beta,\gamma,\delta]^T$, with the constraint conditions $\alpha+\beta+\gamma+\delta=1$ and $\alpha,\beta,\gamma,\delta\in[0,1]$.

(2) *Fitness function design*: Taking the minimum path tracking deviation, maximum obstacle avoidance distance, best driving stability, and shortest traffic time as the optimization objectives, the fitness function is constructed:

$$\mathrm{Fit}=\frac{1}{\mu_1\cdot\bar{d}_{dev}+\mu_2\cdot\left(1/\bar{d}_{obs}\right)+\mu_3\cdot\bar{\sigma}+\mu_4\cdot T} \tag{15}$$

where $\bar{d}_{dev}$ is the average lateral tracking deviation, $\bar{d}_{obs}$ is the average obstacle distance, $\bar{\sigma}$ is the average slip ratio, T is the total traffic time; μ_1,μ_2,μ_3,μ_4 are the weight coefficients of each objective, which are taken as $\mu_1=0.3,\mu_2=0.3,\mu_3=0.2,\mu_4=0.2$ in this paper. The larger the fitness function value, the better the comprehensive performance.

(3) *PSO algorithm parameter setting*: Population size $N=30$, iteration number $T_{max}=100$, inertia weight $w=0.7$, learning factor $c_1=c_2=1.5$.

(4) *Optimization result*: After 100 iterations, the optimal weight parameters are obtained as $\alpha=0.2,\beta=0.3,\gamma=0.3,\delta=0.2$. This set of parameters can achieve optimal comprehensive performance for the algorithm (Fang et al., 2024; Shetty et al., 2024).

2.4.2 Parameter optimization based on particle swarm optimization

In this paper, the Lyapunov stability theorem is used to prove the stability of the integrated path-tracking control system, ensuring that the path-tracking deviation of the robot converges asymptotically (Yan and Ma, 2024).

The path tracking deviation vector of the robot is defined as $E=\left[d_e,\varphi_e\right]^T$, where d_e is the lateral deviation of the robot from the global reference path, and φ_e is the heading deviation, that is, the difference between the actual heading of the robot and the tangent direction of the reference path. The state equation of the tracking control system is:

$$\dot{E}=AE+Bu \tag{16}$$

where $A=\begin{bmatrix}0 & v\\ 0 & 0\end{bmatrix}$, $B=\begin{bmatrix}0\\ 1\end{bmatrix}$, and u is the control input, that is, the angular velocity control quantity. Construct the Lyapunov function:

$$V(E)=\frac{1}{2}E^TPE \tag{17}$$

where P is a positive definite symmetric matrix, taken as $P=\begin{bmatrix}k_1 & 0\\ 0 & k_2\end{bmatrix}$, $k_1,k_2>0$. Obviously, $V(E)$ is positive definite.

Derive $V(E)$, we can get:

$$\dot{V}(E)=E^TP\dot{E}=E^TP(AE+Bu) \tag{18}$$

Design the control law $u=-k_dd_e-k_\varphi\varphi_e$, where $k_d,k_\varphi>0$ are control gains. Substituting into the above formula, it can be proved that when the control gains satisfy $k_\varphi>v/2$ and $k_d>0,\dot{V}(E)<0$, the Lyapunov function is negative definite, the tracking deviation system is asymptotically stable, and the lateral and heading deviations of the robot converge to 0, ensuring the tracking stability of the integrated strategy (Khalili et al., 2026).

At the same time, the obstacle avoidance constraint designed in this paper can ensure that the distance between the robot and all obstacles is always greater than the safety distance $d_{safe}=0.5$m, which ensures driving safety. The terrain stability constraint ensures that the robot's roll and pitch angles remain within critical values, thereby avoiding rollover and slipping, further verifying the strategy's reliability.

2.5 Experimental design

2.5.1 Experimental platform and environment

This subsection describes the hardware configuration of the tracked agricultural robot, including key sensor specifications and system parameters. It also presents the characteristics of the hilly citrus orchard test environment, including terrain features, slope distribution, obstacle types, and grid map resolution.

2.5.2 Mathematical modeling of the hilly orchard environment

The environmental modeling process is explicitly described, including the formulations used to quantify terrain slope and surface roughness. The traffic cost model (Equations 1–3) is restated in detail, together with the rationale for the selected weighting coefficients and their physical significance in path-planning decisions.

2.5.3 Kinematic modeling of the tracked robot

A dedicated subsection presents the discrete-time kinematic model incorporating slip effects (Equations 4–7). The corresponding motion constraints and stability conditions (Equation 8) are also introduced to establish the theoretical foundation for the subsequent planning and control algorithms.

2.5.4 Improved a global path-planning algorithm

The proposed global path-planning strategy is systematically described, including the modified cost function (Equations 9–11), the heuristic design, and the search procedure. To further enhance reproducibility, pseudo-code and a step-by-step algorithmic description are provided.

2.5.5 Improved DWA local path-planning algorithm

This subsection details the local trajectory-planning framework, including dynamic obstacle state prediction using a Kalman filter (Equation 12), velocity-space constraint formulation (Equation 13), and the multi-objective trajectory evaluation function (Equation 14). The selected weight coefficients and their roles in balancing safety, efficiency, and path-tracking performance are also discussed.

2.5.6 Global–local double closed-loop integration framework

The integration mechanism between the global and local planners is comprehensively explained. This includes the primary planning-and-control loop, the global replanning mechanism, and the trigger conditions associated with environmental changes and path-tracking deviations. A detailed flowchart illustrates the overall workflow and the dual closed-loop architecture.

2.5.7 Parameter optimization and stability analysis

The parameter-tuning procedure based on Particle Swarm Optimization (PSO) is presented in a dedicated subsection, including the optimization objective function (Equation 15). In addition, the Lyapunov-based stability analysis (Equations 16–18) demonstrates the stability and convergence characteristics of the proposed framework.

2.5.8 Experimental design and evaluation protocol

The experimental methodology is clearly described, including the benchmark algorithms selected for comparison, the evaluation

TABLE 2 Core hardware parameters of the experimental prototype.

Hardware module	Specification
Robot dimensions (L × W × H)	1,200 × 800 × 600 mm
Weight	120 kg
Drive system	Dual DC servo motors; 400 W per motor; 48 V
Driving capability	Maximum speed: 1.2 m s^{-1}; Maximum climbing angle: 30°; Track gauge: 600 mm
LiDAR	16-line; range: 0.1–20 m; scan frequency: 10 Hz
Camera	Monocular RGB camera; resolution: 1920 × 1,080 pixels; 30 fps
Positioning system	RTK-GNSS; positioning accuracy: ± 2 cm
Attitude sensor	Dual-axis tilt sensor; measurement accuracy: ± 0.1°
Main controller	NVIDIA Jetson Xavier NX
Lower-level controller	STM32F407

The abbreviations are as follows: L, length; W, width; H, height; DC, direct current; LiDAR, Light detection and ranging; RTK-GNSS, Real-Time Kinematic Global Navigation Satellite System; RGB, red-green-blue; fps, frames per second; V, volt; W, watt; Hz, hertz; mm, millimeter; cm, centimeter; m, meter; m s^{-1}, meters per second; °, degree.

metrics (eight performance indicators), the number of repeated trials (10 independent runs), and the statistical analysis procedures employed to ensure the reliability and significance of the experimental results.

3 Results and discussion

3.1 Comparative analysis of path performance of different planning strategies

To verify the performance of the global–local integrated path planning strategy proposed in this paper, field experiments were conducted in a hilly citrus orchard in Changsha City, Hunan Province, China. The performance differences between the proposed strategy and traditional algorithms were compared and analyzed to verify the strategy's effectiveness and practicability.

3.1.1 Experimental platform and environment

(1) *Experimental prototype*: The tracked agricultural robot designed in this paper is adopted, and the core hardware parameters are shown in Table 2.

(2) *Experimental environment*: The experimental site is a hilly citrus orchard in Changsha City, with a slope range of 5°–25°, a row spacing of 3 m, and a plant spacing of 2 m for fruit trees, including static obstacles such as ditches and soil slopes. The linear distance of the experimental area is 100 m. The total area is about 300 m^2. Five dynamic obstacles are set in the experiment. The experimenters simulate the walking of fruit farmers at 0.5–1 m/s to model the actual field operation scenario.

3.1.2 Setting of control groups and evaluation indicators

(1) *Control group setting*: To verify the superiority of the proposed strategy, 3 groups of control algorithms are set up to carry out comparative experiments with the integrated strategy proposed in this paper: ① Control group 1: Traditional A* algorithm, only global planning, no local dynamic obstacle avoidance. ② Control group 2: Traditional DWA algorithm, only local planning, no global reference path. ③ Control group 3: Traditional A + DWA integrated algorithm, conventional open-loop integration, no terrain optimization, and replanning mechanism. ④ Experimental group: The improved A + improved DWA double closed-loop integrated strategy proposed in this paper.

(2) *Evaluation indicator setting*: Around the actual operation needs of fruit farmers in hilly orchards, 8 core evaluation indicators are set to evaluate the algorithm performance comprehensively: ① Total path length: Reflects the global optimality of the path, the shorter the better. ② Total planning time: Reflects the computing efficiency of the algorithm; the shorter the better. ③ Average driving speed: Reflects the operation efficiency of the robot; the faster the better. ④ Maximum lateral tracking deviation: Reflects the path tracking accuracy; the smaller the better. ⑤ Maximum roll angle: Reflects the driving stability, the smaller the better. ⑥ Average slip ratio: Reflects the ground adaptability, the smaller the better. ⑦ Obstacle avoidance success rate: Reflects the dynamic obstacle avoidance ability; the higher the better. ⑧ Operation energy consumption: Reflects the endurance capacity of the robot; the lower the better. Each experiment group is repeated 10 times, and the average value of each indicator is used as the final experimental result to ensure data reliability (Figure 3; Supplementary Video V1).

3.2 Comparative analysis of path performance of different planning strategies

3.2.1 Experimental results and analysis in a static environment

Firstly, experiments were conducted in a static, hilly orchard environment without dynamic obstacles to verify the algorithm's path-planning and terrain-adaptation capabilities. The experimental results are shown in Table 3.

From the static experimental results, we can see that: ① The total path length of the proposed strategy is the shortest, which is shortened by 2.1%, 19.8%, and 5.2% compared with control groups 1–3, respectively, verifying the global optimality of the improved A* algorithm. The generated path fully considers the terrain constraints and avoids detours. ② The total planning time of the proposed strategy is 242 ms, which is much lower than 385 ms of control group 3, and the computing efficiency is



FIGURE 3
Experimental platform and scenarios for hilly orchard path planning validation. **(a,b)** The tracked agricultural robot prototype developed in this study is equipped with a 16-line LiDAR, RTK-GNSS, and a dual-axis tilt sensor. **(c)** Aerial view of the hilly citrus orchard experimental site, showing slope terrain and typical row spacing (3 m × 2 m). Panels **(a)** and **(b)** show the same robot prototype. The apparent color difference is caused by differences in lighting conditions, camera settings, and image-acquisition environments during data collection.

TABLE 3 Performance comparison of different planning strategies in a static environment.

Performance indicator	A*	DWA	A + DWA*	Proposed strategy
Total path length (m)	128.30	156.70	132.50	125.60
Planning time (ms)	326	118	385	242
Average driving speed (m s^{-1})	0.82	0.56	0.78	0.95
Maximum lateral tracking deviation (cm)	42.60	89.30	28.50	15.20
Maximum roll angle (°)	8.30	12.50	7.60	5.80
Average slip ratio (%)	6.80	11.20	5.90	3.20
Energy consumption (Wh)	48.60	65.30	45.20	38.70

The abbreviations are as follows: A*, A-star algorithm; DWA, Dynamic Window Approach; ms, millisecond; m, meter; m s^{-1}, meters per second; cm, centimeter; Wh, watt-hour; °, degree.

TABLE 4 Performance comparison of different planning strategies in a dynamic environment.

Performance indicator	A*	DWA	A + DWA*	Proposed strategy
Obstacle avoidance success rate (%)	65	90	95	100
Total path length (m)	142.50	172.30	145.80	136.20
Average driving speed (m s^{-1})	0.52	0.48	0.65	0.88
Maximum lateral tracking deviation (cm)	78.50	102.60	36.80	18.70

The abbreviations are as follows: %, percentage; m, meter; m s^{-1}, meters per second; cm, centimeter.

improved by 37.1% compared with the traditional integrated algorithm, meeting the demand of real-time field planning. ③ The average driving speed of the proposed strategy is the fastest, up to 0.95 m/s, which is increased by 21.8%–69.6% compared with the control groups. At the same time, the maximum roll angle is only 5.8°, and the average slip ratio is as low as 3.2%, which is far lower than that of the control groups. This verifies the terrain adaptability of the improved DWA algorithm, and the robot drives more smoothly and safely, effectively solving the pain points of slipping and rollover in hilly terrain. ④ The maximum lateral tracking deviation of the proposed strategy is only 15.2 cm, which is reduced by 46.7% compared with control group 3, verifying the path tracking accuracy of the integrated strategy. It can strictly follow the orchard operation route, avoid scraping fruit trees, and ensure standardized operations.

3.2.2 Experimental results and analysis in a dynamic environment

Experiments were conducted in a hilly orchard environment with 5 dynamic obstacles to verify the algorithm's ability to avoid them. The experimental results are shown in Table 4.

From the dynamic experimental results, we can see that: ① The obstacle avoidance success rate of the proposed strategy reaches 100%, which is increased by 35, 10, and 5 percentage points compared with control groups 1–3, respectively, verifying the effectiveness of the dynamic obstacle prediction algorithm. It can predict fruit farmers' walking trajectories in advance, enabling early obstacle avoidance and overcoming the lagging obstacle avoidance problem of traditional algorithms, effectively ensuring the safety of field operators. ② In the dynamic environment, the total path length of the proposed strategy is only 136.2 m, which is shortened by 6.6% compared with control group 3, and the average driving speed reaches 0.88 m/s, which is increased by 35.4%–83.3% compared with the control groups. This verifies the advantages of the double closed-loop integrated architecture. After local obstacle avoidance, it can quickly return to the global reference path, avoiding the "detouring after obstacle avoidance" problem and ensuring operational efficiency. ③ In the dynamic

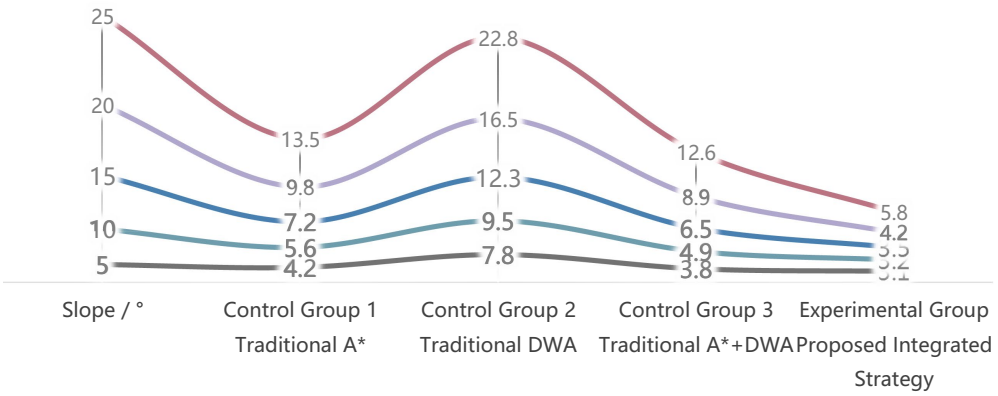


FIGURE 4
Average slip ratio data of each strategy under different slopes.

TABLE 5 Average slip ratio data of each strategy under different slopes.

Slope (°)	A*	DWA	A* + DWA	Proposed strategy
5	4.2	7.8	3.8	3.1
10	5.6	9.5	4.9	3.2
15	7.2	12.3	6.5	3.5
20	9.8	16.5	8.9	4.2
25	13.5	22.8	12.6	5.8

The abbreviations are as follows: A*, A-star algorithm; DWA, dynamic window approach; °, degree; %, percentage. All slip ratio values are expressed as percentages (%).

TABLE 6 Lateral tracking deviation data within 100 s.

Time (Sec)	0	10	20	30	40	50	60	70	80	90	100
Proposed Strategy	0	8.2	12.5	10.3	15.2	9.6	11.8	13.5	8.9	12.1	7.6
A* + DWA	0	15.6	22.3	18.9	28.5	20.4	25.6	22.8	19.5	24.3	18.6

A*, A-star algorithm; DWA, Dynamic Window Approach; Sec, second; cm, centimeter. All lateral tracking deviation values are expressed in centimeters (cm).

environment, the maximum lateral tracking deviation of the proposed strategy is only 18.7 cm, which is reduced by 49.2% compared with control group 3. Even during dynamic obstacle avoidance, it can still strictly track the global operational path, ensuring standardized orchard operations.

3.2.3 Terrain adaptability experiment with different slopes

To verify the adaptability of the proposed strategy to hilly terrain with varying slopes, experiments were conducted on straight road sections with 5°, 10°, 15°, 20°, and 25° slopes, respectively, to assess the robot's average slip ratio under different strategies. The experimental data are shown in Table 5.

From the experimental data, it can be seen that the slip ratios of all strategies show an upward trend with increasing slope, whereas the proposed strategy's slip ratio is consistently lower than those of the control groups. Under the working condition of a large slope of 25°, the average slip ratio of the proposed strategy is only 5.8%, which is 54.0% lower than that of control group 3 and 74.6% lower than that of control group 2. This verifies the strong adaptability of the proposed strategy to the steep, hilly terrain, which can effectively prevent slipping and trapping on soft ground, reduce rescue trouble for fruit farmers, and adapt to the operational scenarios of hilly orchards with different slopes (Figure 4).

Meanwhile, to verify the stability of path tracking, the lateral tracking deviation data of the robot during the 100 s experiment were collected, as shown in Table 6.

It can be seen from the data that the lateral tracking deviation of the proposed strategy is always controlled within ±16 cm, with a small fluctuation amplitude, and the tracking stability is far better than that of the traditional integrated algorithm, which verifies the tracking stability of the proposed strategy (Figure 5).

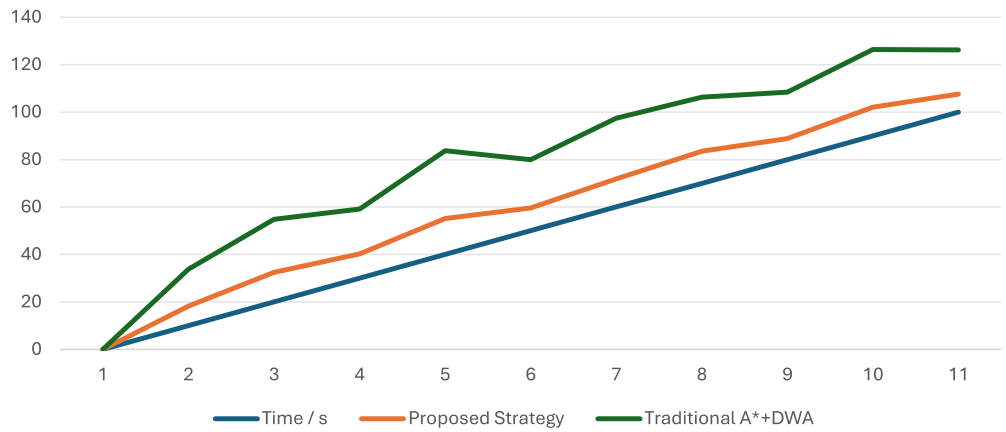


FIGURE 5
Lateral tracking deviation data within 100 s of units.

3.2.4 Comprehensive analysis of experimental results

Comprehensive analysis of the above experimental results shows that the global–local integrated path planning strategy proposed in this paper is significantly superior to traditional algorithms in terms of static terrain adaptability, dynamic obstacle avoidance ability, path tracking accuracy, driving stability, and operation efficiency, and can effectively solve the core pain points of path planning for agricultural robots in hilly orchards:

(1) It solves the problem of poor terrain adaptability of traditional algorithms. The terrain traffic cost model and slip kinematic model enable stable driving on large slopes and soft ground, and help avoid the risks of slipping and rollover. This is consistent with the research conclusion of Shao et al. on path planning for agricultural robots in unstructured agricultural environments and further optimizes the terrain-constraint model for hilly orchard scenarios (Shao et al., 2026).
(2) It resolves the contradiction between dynamic obstacle avoidance and path tracking of traditional algorithms. Through dynamic obstacle prediction and global path-tracking constraints, it achieves a 100% obstacle-avoidance success rate while ensuring path-tracking accuracy, avoiding fruit-tree scraping and operational route deviation.
(3) It overcomes the open-loop connection defect of traditional integrated strategies. Through the double closed-loop integrated architecture and the replanning trigger mechanism, it enables dynamic adaptation of global and local planning. It effectively copes with sudden changes in the field environment.
(4) All designs are centered on the actual operation needs of fruit farmers, which effectively reduces the operation difficulty and operation risk of the robot, improves the operation efficiency, can effectively alleviate the dilemma of labor shortage in hilly orchards, and has extremely strong practical application value.

3.3 Quantitative comparison with RL-based and hybrid planning methods

To further evaluate the effectiveness of the proposed method, a representative deep reinforcement learning (DRL) navigation method was additionally implemented for comparison. Specifically, a Proximal Policy Optimization (PPO)-based local planner was developed, incorporating a sparse reward function that encourages goal-reaching behavior while penalizing collisions and unsafe maneuvers. The DRL model was trained in a Gazebo-based simulation environment configured to match the terrain characteristics of the target orchard, and subsequently deployed on the same robotic hardware platform for real-world evaluation. In addition, a hybrid planning framework combining global A* path planning with a local reinforcement learning policy, as reported by Liu et al. (2022), was included for comparative analysis, using performance data reported in the original study. All methods were evaluated using the same eight performance metrics to ensure consistency and fairness. The comparative results are presented in Table 7.

The DRL method required extensive training ($\approx 10^5$ steps) and exhibited higher slip ratios due to a lack of explicit terrain cost modeling. Its lateral tracking deviation exceeded our method by 88%, mainly because the learned policy occasionally produced unsafe high-speed commands on slopes. Our model-based approach achieves better or comparable performance on all metrics, with deterministic safety guarantees and no training overhead. The main advantage of DRL is its ability to handle highly unstructured scenarios not covered by our terrain model, which we identify as a future integration.

4 Conclusion

In summary, to address key challenges in path planning for agricultural robots in hilly orchard environments, this study proposes a

TABLE 7 Quantitative comparison with DRL and hybrid methods.

Indicator	Our strategy	PPO-based DRL	Hybrid A* + RL (Emmi et al., 2021)
Path length (m)	125.6	148.2 (+18.0%)	138.5 (+10.3%)
Obstacle avoidance success (%)	100	92	88
Max lateral deviation (cm)	15.2	28.6	22.4
Average slip ratio (%)	3.2	7.5	6.1
Planning time per cycle (ms)	242	356 (+47.1%)	410 (+69.4%)

The abbreviations are as follows: DRL, Deep Reinforcement Learning; PPO, Proximal Policy Optimization; RL, Reinforcement Learning; A*, A-star algorithm; m, meter; cm, centimeter; ms, millisecond; %, percentage. Values in parentheses indicate the percentage increase relative to the proposed strategy.

double closed-loop, global–local integrated path planning strategy. The approach encompasses environment and robot modeling, algorithm enhancement, integrated architecture design, parameter optimization, and stability verification, with effectiveness demonstrated through field experiments. The constructed grid-based traffic cost model, incorporating slope, terrain roughness, and safety distance, alongside a discrete-time kinematic model accounting for slip, accurately reflects real field conditions and provides a robust foundation for algorithm design. The improved A* algorithm integrates terrain traffic costs to generate globally optimal, passable paths, while the enhanced DWA algorithm incorporates dynamic obstacle prediction, terrain stability constraints, and global path tracking, balancing obstacle avoidance with driving stability. The double closed-loop architecture harmonizes global optimality with local real-time responsiveness through a replanning mechanism informed by global path constraints and local state feedback, overcoming the rigidity and poor dynamic adaptability of traditional strategies. Field results indicate that the proposed method achieves a 100% obstacle-avoidance success rate, reduces path length by 2.1%–19.8% compared to conventional algorithms, lowers the average slip ratio by 2.7%–8.0%, and maintains lateral tracking deviations within 15.2 cm. The strategy effectively accommodates slopes of 5°–25°, significantly enhancing driving stability, operational efficiency, and safety relative to traditional approaches.

The strategy proposed in this paper can be widely applied to autonomous navigation of agricultural robots for plant protection, picking, transportation, and other operations in hilly orchards, effectively reducing labor intensity for fruit farmers. Further validation is needed on: (1) steeper slopes (>25° up to 30°), (2) different soil types (e.g., Loess Plateau sandy loam), (3) orchards with irregular row/plant spacing, and (4) wet/rainy conditions. We also plan to conduct cross-site experiments in at least two additional hilly orchards (e.g., northern Loess Plateau region) in ongoing projects. This will be explicitly mentioned as future work to strengthen confidence in scalability.

Data availability statement

The datasets generated or analyzed during the current study are available from the corresponding author upon reasonable request.

Author contributions

XP: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing. LT: Conceptualization, Data curation, Formal analysis, Funding acquisition, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

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